Quadratics & Polynomials

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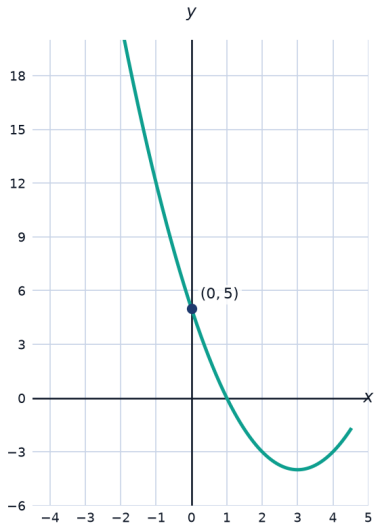
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Factoring quadratics**

Solve $x^2 + 7x + 12 = 0$ by factoring.

2 **[Foundational]** • **Target: Vertex form and completing the square**

Write $y = x^2 - 6x + 5$ in vertex form by completing the square, and state the vertex.



3 **[Foundational]** • **Target: Vertex form and completing the square**

Solve $x^2 + 8x + 3 = 0$ by completing the square.

4 **[Foundational]** • **Target: The quadratic formula and discriminant**

Solve $3x^2 - 5x - 2 = 0$ using the quadratic formula.

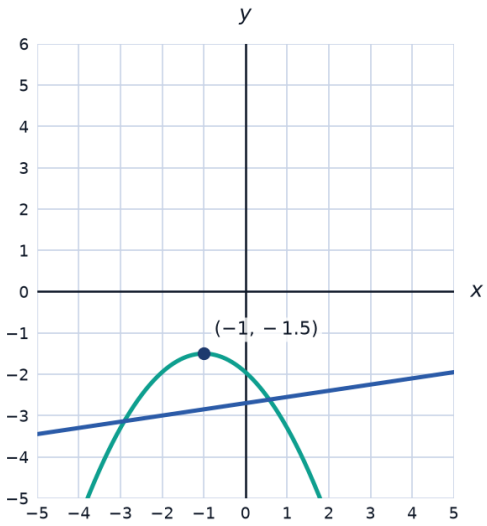
5 **[Foundational]** • **Target: The quadratic formula and discriminant**

Find the discriminant of $2x^2 + 3x + 5 = 0$ and state the nature of its roots.

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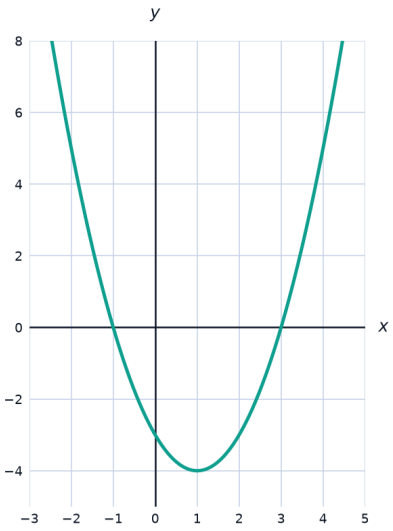
6 [Foundational] • Target: The quadratic formula and discriminant

For what values of k does $x^2 + kx + 16 = 0$ have exactly one repeated real root?



7 [Foundational] • Target: Graphing parabolas

The graph of $y = x^2 - 2x - 3$ is shown below. Use the graph to state the x-intercepts, then confirm your answer algebraically by factoring.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Graphing parabolas

The graph of $y = -2x^2 + 8x - 3$ opens which direction, and what is the x-coordinate of its vertex?

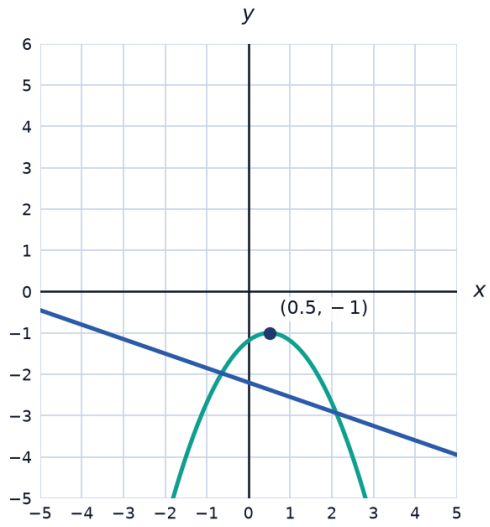
9 [Intermediate] • Target: Polynomial division and the remainder theorem

Use synthetic division to divide $x^3 - 2x^2 - 5x + 6$ by $(x - 1)$, and state the quotient and remainder.

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10 [Intermediate] • Target: Polynomial division and the remainder theorem

Use the Remainder Theorem to find the remainder when $x^3 + 4x^2 - x + 2$ is divided by $(x + 2)$.



11 [Intermediate] • Target: Quadratic word problems

A ball is thrown upward so that its height is modeled by $h(t) = -16t^2 + 64t + 5$ (in feet), where t is time in seconds. Find the time at which the ball reaches its maximum height, and find that maximum height.

12 [Intermediate] • Target: Quadratic word problems

The area of a rectangle is $x^2 + 2x - 15$ square meters, and its width is $x - 3$. Find its length in terms of x by factoring.

13 [Intermediate] • Target: Sum and product of roots

For $x^2 - 7x + 10 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.

14 [Intermediate] • Target: Sum and product of roots

A quadratic equation has roots 3 and -5. Write the equation in standard form.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

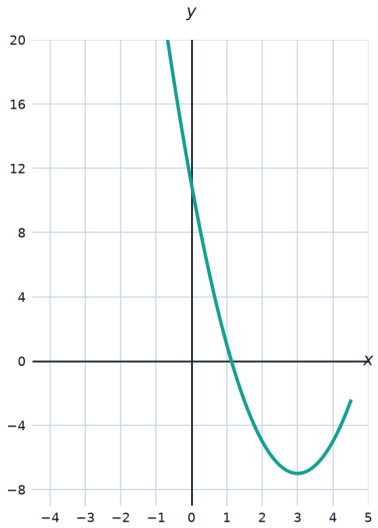
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15 [Challenging] • Target: A well-designed word problem: maximizing weekly profit • Cross-Lesson: 3.01 Functions & Graphs

This question has three parts. A company's weekly profit (in thousands of dollars) from producing x hundred units is modeled by $P(x) = -2x^2 + 20x - 32$. (a) Factor $P(x)$ to find the break-even production levels, where profit is zero. (b) Find the vertex of the parabola to determine the production level that maximizes profit, and state the maximum profit. (c) State the range of production levels (in hundreds of units) for which profit is positive.

16 [Challenging] • Target: Quadratic word problems • Cross-Lesson: 4.01 Ratios, Rates & Data

Convert the quadratic function $y = 2x^2 - 12x + 11$ into vertex form $y = a(x - h)^2 + k$. State the coordinates of the vertex and the line of symmetry.



17 [Challenging] • Target: Sum and product of roots • Cross-Lesson: 5.01 Statistics & Probability

A parabola has its vertex at $(3, -4)$ and passes through the point $(1, 0)$. (a) Write the equation of the parabola in standard form $y = ax^2 + bx + c$. (b) Determine the coordinates of the other x-intercept.

18 [Challenging] • Target: A well-designed word problem: maximizing weekly profit • Cross-Lesson: 6.01 Geometry

If $x - 2$ is a factor of the polynomial $P(x) = x^3 - kx^2 + 3x - 10$, find the value of k, and determine all real roots of $P(x)$.

19 [Challenging] • Target: Factoring quadratics • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Solve the quadratic equation $3x^2 + 10x - 8 = 0$ by factoring.

20 [Challenging] • Target: Vertex form and completing the square • Cross-Lesson: 8.01 Exponentials & Logarithms

Use the quadratic formula to solve $x^2 + 2x - 8 = 0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Factoring quadratics

Find the vertex of the parabola $y = -2x^2 - 28x - 92$.

a. $(-7, 6)$

b. $(-5, 6)$

c. $(7, 6)$

d. $(-7, -6)$

22 [Foundational] • Target: Vertex form and completing the square

What is the sum of the roots of the quadratic equation $3x^2 - 11x + 2 = 0$?

a. $-\frac{2}{3}$

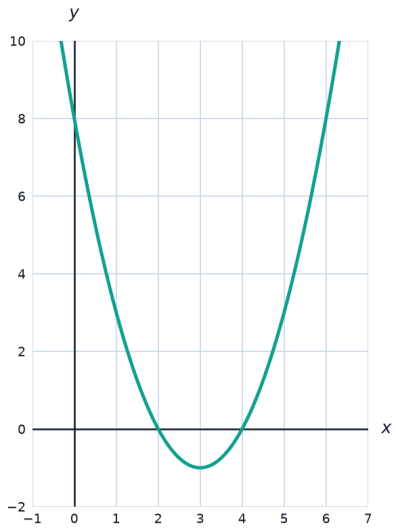
b. $-\frac{11}{3}$

c. $\frac{11}{3}$

d. $\frac{2}{3}$

23 [Foundational] • Target: The quadratic formula and discriminant

The graph of the quadratic function $y = x^2 - 6x + 8$ is shown below. What are the coordinates of the vertex of the parabola?



a. (3, -1)

b. (3, 1)

c. (1, 3)

d. (-1, 3)

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24 [Foundational] • Target: Graphing parabolas

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The parabola has its vertex at $(-2, 8)$ and passes through the point $(0, 4)$. Determine the value of c.

a. 4

b. -2

c. 8

d. 2

25 [Foundational] • Target: Polynomial multiplication and special products

The graph of the quadratic function $y = x^2 - 6x + 9$ is a parabola. What is the x-coordinate of the vertex of this parabola?

a. 3

b. -3

c. 0

d. 6

26 [Foundational] • Target: Polynomial division and the remainder theorem

When polynomial $P(x) = 2x^3 - 3x^2 + 5x - 7$ is divided by $(x - 1)$, what is the remainder?

a. -3

b. 3

c. -7

d. 0

27 [Foundational] • Target: Quadratic word problems

For $2x^2 + 14x + 9 = 0$, what is the sum of the roots?

a. 7

b. -7

c. -6

d. 9/2

28 [Foundational] • Target: Sum and product of roots

What is the product of the solutions to the quadratic equation $3x^2 + 11x - 12 = 0$?

a. -4

b. 4

c. $-\frac{11}{3}$

d. -12

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: A well-designed word problem: maximizing weekly profit**

What is the sum of the solutions to the quadratic equation $2x^2 - 8x + 5 = 0$?

a. 4

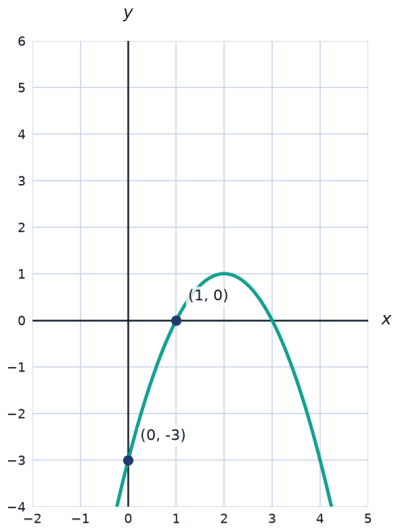
b. -4

c. $\frac{5}{2}$

d. 8

30 **[Intermediate]** • **Target: Factoring quadratics**

The graph of the quadratic function $y = ax^2 + bx + c$ is shown below. The parabola intersects the x-axis at points $(1, 0)$ and $(p, 0)$, and has a y-intercept of $(0, -3)$. If the axis of symmetry of the parabola is the line $x = 2$, find the value of p.



a. 3

b. 4

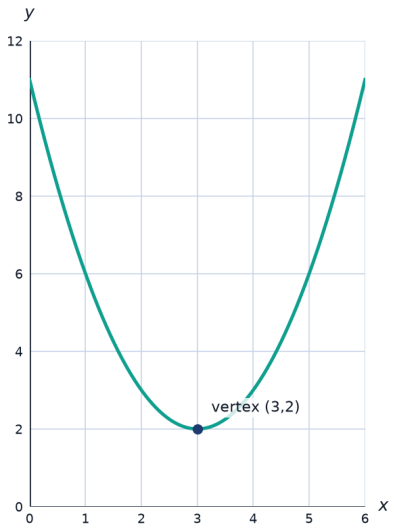
c. 5

d. 6

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31 **[Intermediate]** • **Target: Vertex form and completing the square**

What is the vertex of the parabola $y = x^2 - 6x + 11$?



a. $(3, 2)$

b. $(-3, 2)$

c. $(3, -2)$

d. $(6, 11)$

32 **[Intermediate]** • **Target: The quadratic formula and discriminant**

What are the solutions to the quadratic equation $x^2 - 10x + 20 = 0$?

a. $10 \pm \sqrt{5}$

b. 5 ± 5

c. $-5 \pm \sqrt{5}$

d. $5 \pm \sqrt{5}$

33 **[Intermediate]** • **Target: Graphing parabolas**

If $(x - 3)$ is a factor of the polynomial $P(x) = 2x^3 - kx^2 + 1x - 3$, what is the value of the constant k?

a. 6

b. 3

c. 8

d. -6

34 **[Intermediate]** • **Target: Polynomial multiplication and special products**

The quadratic function $f(x) = 2x^2 - 12x + 23$ is rewritten in vertex form $f(x) = a(x - h)^2 + k$. What is the minimum value of $f(x)$?

a. 5

b. 3

c. 23

d. 11

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35 [Intermediate] • Target: Polynomial division and the remainder theorem

If $x = 3$ is a solution to the quadratic equation $2x^2 + kx - 15 = 0$, what is the value of the constant k?

a. -1

b. 1

c. -3

d. 3

36 [Intermediate] • Target: Quadratic word problems

Consider the quadratic function $f(x) = x^2 - 6x + 9$. What is the value of $f(3)$?

a. 0

b. 3

c. 9

d. 18

37 [Intermediate] • Target: Sum and product of roots

Simplify $\frac{3+4i}{1-2i}$ into Cartesian form $a + bi$.

a. $-1 + 2i$

b. $1 + 2i$

c. $-1 - 2i$

d. $2 + i$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: maximizing weekly profit • Cross-Lesson: 6.01 Geometry

What is the product of the solutions of $x^3 - 4x^2 - 7x + 10 = 0$?

a. -10

b. -12

c. 14

d. 5

39 [Challenging] • Target: Factoring quadratics • Cross-Lesson: 7.01 Trigonometry & Additional Topics

What is the sum of the solutions to $|x^2 - 7| = 9$?

a. 0

b. 8

c. 4

d. 16

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40 [Challenging] • Target: Vertex form and completing the square • Cross-Lesson: 8.01 Exponentials & Logarithms

What is the sum of the solutions of $(a-1)(a^2+8a+15)=0$?

a. 8

b. 7

c. -7

d. -8

41 [Challenging] • Target: The quadratic formula and discriminant • Cross-Lesson: 1.01 Heart of Algebra

What is the vertex of the parabola defined by $y=2x^2-12x+11$?

a. $(3,-7)$

b. $(3,11)$

c. $(-3,-7)$

d. $(6,-7)$

42 [Challenging] • Target: Graphing parabolas • Cross-Lesson: 3.01 Functions & Graphs

If α and β are the roots of $x^2-5x+3=0$, what is the value of $\frac{1}{\alpha}+\frac{1}{\beta}$?

a. $\frac{5}{3}$

b. $\frac{3}{5}$

c. $-\frac{5}{3}$

d. 5

43 [Challenging] • Target: Polynomial multiplication and special products • Cross-Lesson: 4.01 Ratios, Rates & Data

If the polynomial x^4+ax^2+b is divisible by x^2+3x+2, what is $a+b$?

a. -1

b. 1

c. 5

d. -5

44 [Challenging] • Target: Polynomial division and the remainder theorem • Cross-Lesson: 5.01 Statistics & Probability

What is the minimum value attained by the quadratic function $f(x)=3x^2-12x+19$?

a. 7

b. 19

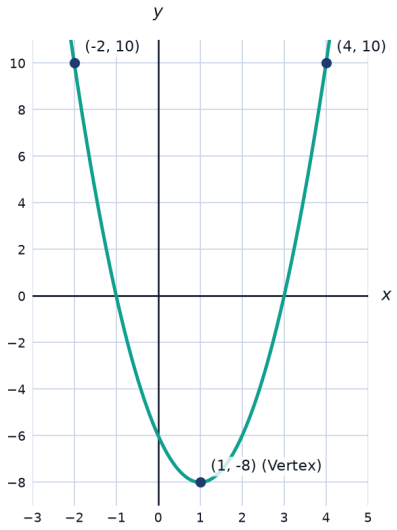
c. 2

d. 3

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45 [Challenging] • Target: Quadratic word problems • Cross-Lesson: 6.01 Geometry

The graph of the quadratic function $g(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(-2, 10)$ and $(4, 10)$, and its vertex is at $(1, -8)$. Determine the value of a.



a. 1

b. 2

c. -1

d. -2

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